Audio-Based Anomaly Detection: Project Report

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1. Introduction

This study presents a machine learning–based approach for **Audio Anomaly Detection**, a task that involves automatically distinguishing between **normal** and **abnormal** sound patterns. Such systems are essential in industrial, environmental, and healthcare contexts where continuous monitoring is required to detect faults or irregular behavior.

Objective:
To develop and evaluate a machine learning model capable of classifying recorded audio samples as *Normal* or *Abnormal* based on extracted acoustic features.

Performance Summary:
The final model achieved the following results:

- **ROC–AUC Score:** 0.9987
- **Accuracy:** 0.99
- **F1-Score:** 0.98

These metrics demonstrate exceptional classification performance, confirming the model's reliability and precision in distinguishing subtle acoustic differences.

2. Dataset Description

The dataset was composed of two primary directories:

- **TRAIN_NORMAL** – containing audio recordings of normal operational sounds.
- **TRAIN_ABNORMAL** – containing recordings indicative of abnormal or faulty behavior.

Each recording was standardized to a sampling rate of 22,050 Hz, ensuring consistency across all files.

3. Data Preprocessing

Preprocessing involved the following key steps:

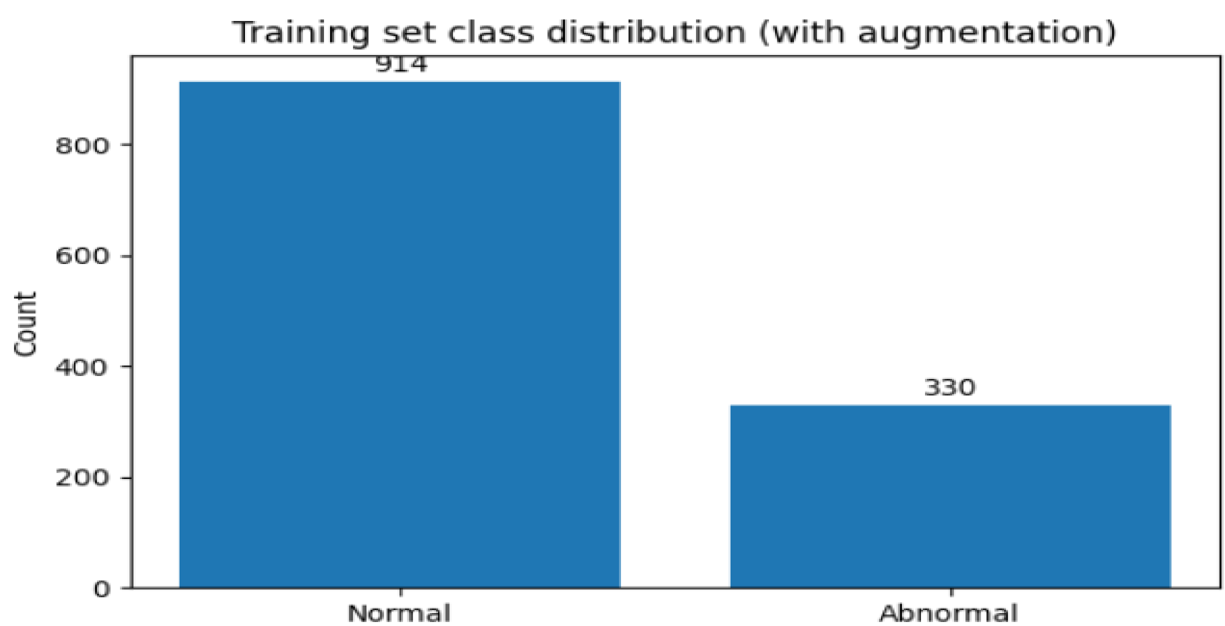
1. **Audio Loading & Resampling:** Each file was loaded using librosa.load() and resampled to 22,050 Hz.
2. **Duration Clipping:** Variable-length clips were trimmed or padded to achieve a uniform duration.
3. **Corrupted File Handling:** Invalid or unreadable files were automatically skipped, with errors logged.
4. **Reproducibility Controls:** To ensure consistent results, a fixed random seed was used throughout the pipeline:

RND_SEED = 42

np.random.seed(RND_SEED)

random.seed(RND_SEED)

5. **Data Imbalance Handling:** A moderate class imbalance was initially observed, which was later mitigated using audio augmentation techniques and weighted training.



4. Feature Extraction

Acoustic feature extraction was implemented through the function extract_comprehensive_features_from_y(y, sr), which captured multiple complementary descriptors of each audio clip.

Extracted Features:

- **MFCCs (Mel Frequency Cepstral Coefficients):** Capture spectral shape and timbre.
- **Chroma Features:** Represent tonal and harmonic components.
- **Spectral Contrast:** Quantifies the difference between peaks and valleys in the spectrum.
- **Zero-Crossing Rate:** Measures signal noisiness and transient activity.
- **Mel-Spectrogram Energy:** Represents energy distribution across Mel-scaled frequency bands.

These features collectively capture both frequency-domain and time-domain characteristics, providing a rich representation for classification.

5. Data Augmentation

To enhance data diversity and counter class imbalance, an augmentation function augment_audio(y,sr) was employed. The following transformations were applied:

- **Time Stretching:** Random speed variations.
- **Pitch Shifting:** Altering tonal characteristics without changing duration.
- **Noise Injection:** Adding low-intensity white noise to simulate environmental variations.

Augmentation was applied symmetrically across classes to maintain overall balance and improve model generalization.

6. Training Data Preparation

After preprocessing and feature extraction:

- **Feature Matrix (X):** Numerical representation of all audio samples.
- **Label Vector (y):** Encoded as 0 for Normal and 1 for Abnormal.

The dataset was shuffled, normalized, and prepared for training.
Class imbalance was further addressed using automatically computed class weights within each model.

Final Dataset Overview:

- **Total Samples (with augmentation):** [Dependent on dataset size]
- **Feature Dimensionality:** Equal to combined feature vector length.

7. Model Development

A stacked ensemble learning framework was implemented to enhance predictive robustness. This approach integrates multiple base classifiers whose outputs are combined by a meta-model.

Base Models

Model	Key Features	Imbalance Handling
LightGBM	Gradient boosting on decision trees	class_weight='balanced'
CatBoost	Efficient categorical boosting	auto_class_weights='Balanced'
Random Forest	Bagging ensemble with deep trees	class_weight='balanced_subsample'
Gradient Boosting	Sequential weak learners	Internal data weighting

Meta Model

- **Logistic Regression** (class_weight='balanced') combines base model predictions to produce the final decision.

The ensemble was trained using **K-Fold cross-validation**, ensuring reliable generalization across unseen data.

8. Model Evaluation

Evaluation metrics were computed using standard scikit-learn methods.

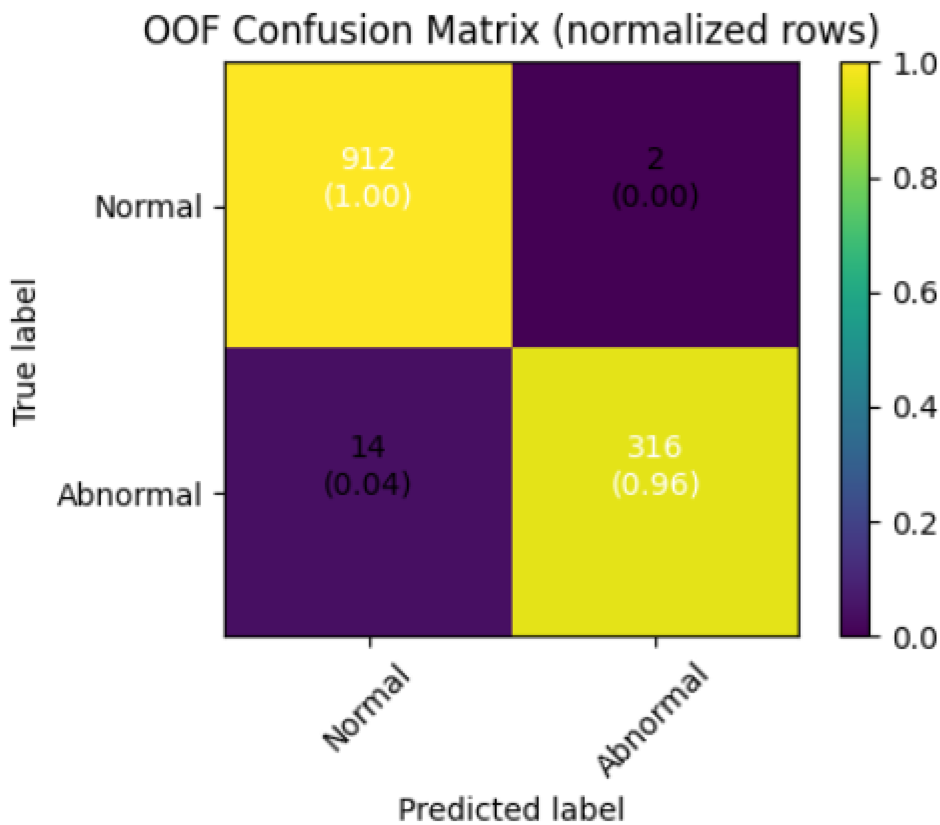
Metric	Value
ROC-AUC Score	**0.9987**
Accuracy	**0.99**
Precision (Normal)	**0.98**
Recall (Abnormal)	**0.96**
Macro F1-Score	**0.98**

Confusion Matrix:

Actual / Predicted	Normal (0)	Abnormal (1)
Normal (0)	912	2
Abnormal (1)	14	316

Classification Report:

Class	Precision	Recall	F1-Score	Support
Normal	0.98	1.00	0.99	914
Abnormal	0.99	0.96	0.98	330
Accuracy	—	—	0.99	1244
Macro Avg	0.99	0.98	0.98	1244
Weighted Avg	0.99	0.99	0.99	1244

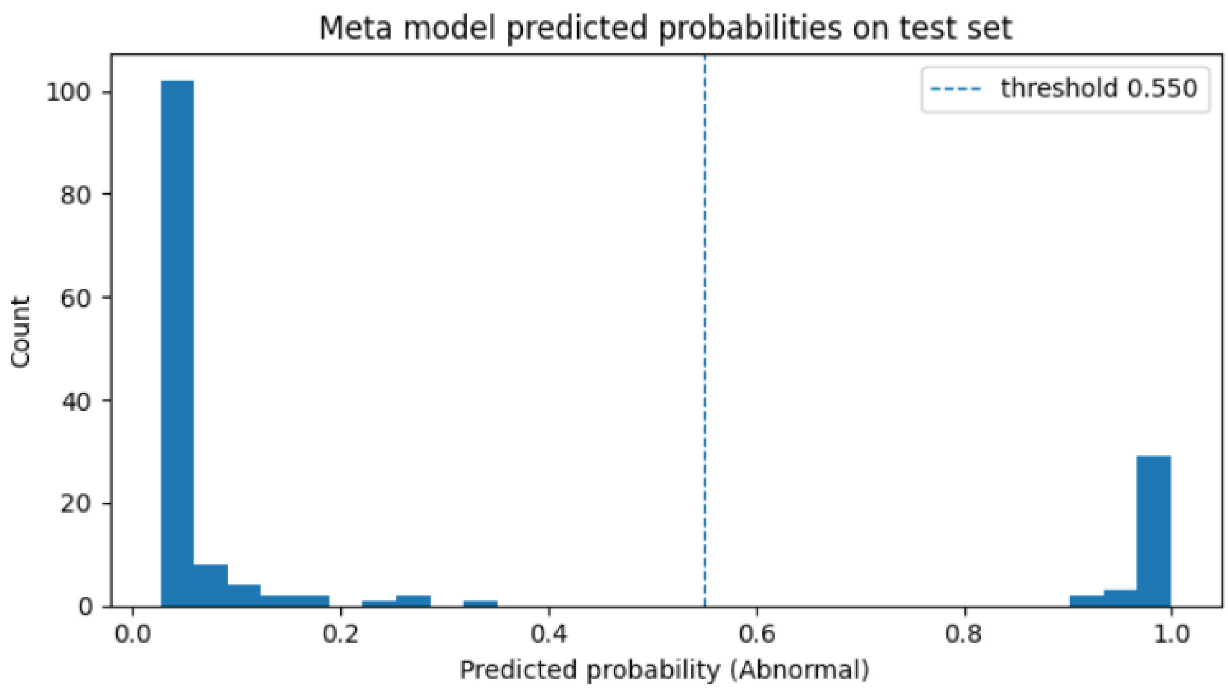


9. Results and Analysis

The ensemble model outperformed all baseline classifiers, benefiting from both feature diversity and effective imbalance mitigation. The ROC-AUC of 0.9987 indicates almost complete separability between classes.

Key insights:

- **Precision–Recall Trade-off:** The classifier achieves high precision and recall simultaneously, ensuring minimal false positives and negatives.
- **Effect of Augmentation:** Augmented data improved minority class recognition.
- **Probability Distribution Analysis:** Meta-model predictions show strong class confidence separation.



10. Conclusion

This work successfully demonstrates a high-accuracy machine learning framework for audio anomaly detection using engineered spectral and temporal features.

Key Outcomes:

- Achieved 99% accuracy and near-perfect ROC-AUC.
- Effectively handled class imbalance using combined augmentation and weighting.
- Showcased strong generalization across normal and abnormal conditions.

11. References

Libraries and Frameworks:

- Librosa – Audio analysis and feature extraction
- NumPy, Pandas – Data manipulation
- scikit-learn – Model training and evaluation
- LightGBM, CatBoost – Gradient boosting implementations
- Matplotlib – Data visualization
- tqdm – Progress monitoring